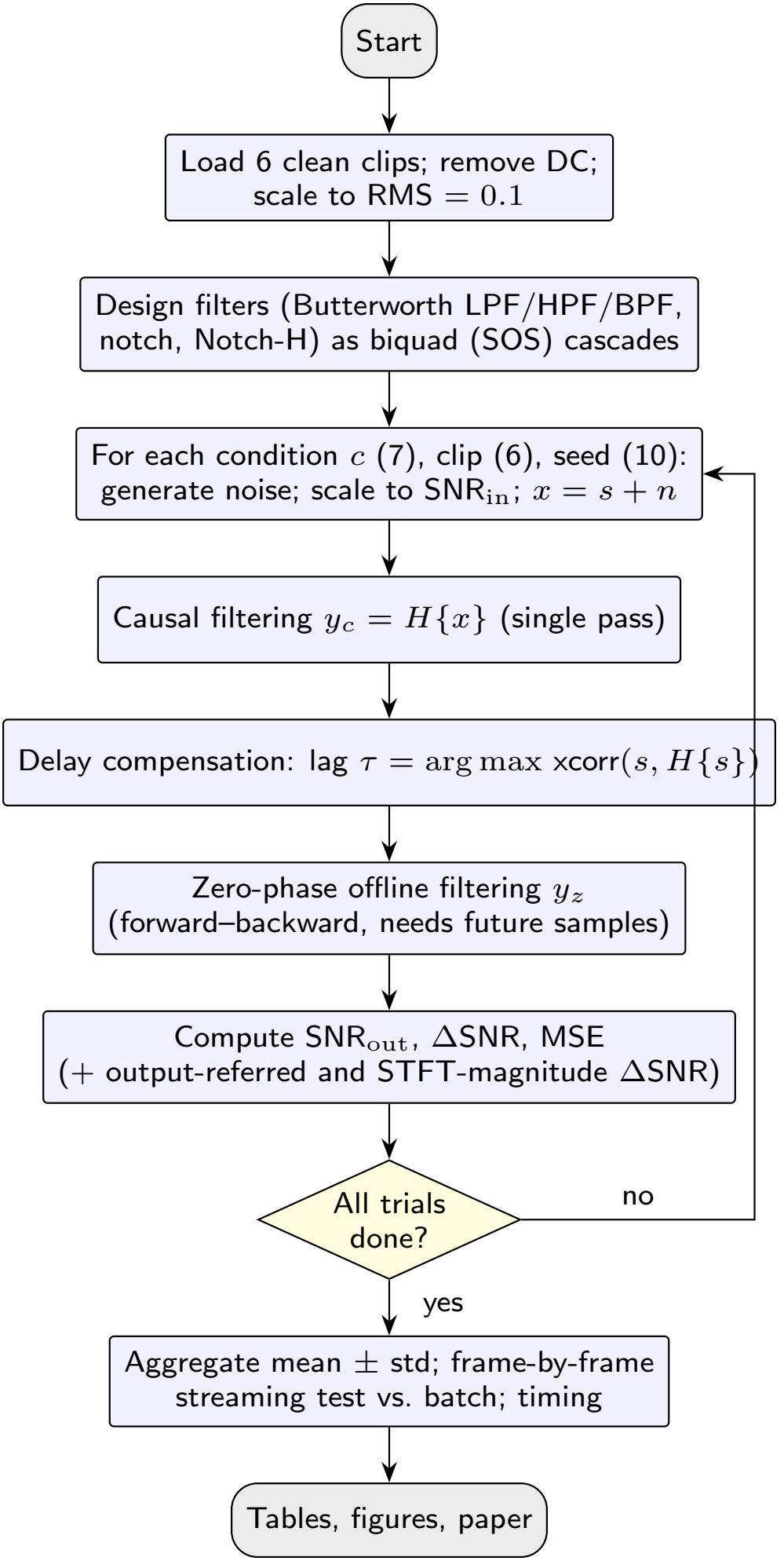

Start
Load 6 clean clips; remove DC; scale to RMS = 0.1
Design filters (Butterworth LPF/HPF/BPF, notch, Notch-H) as biquad (SOS) cascades
For each condition c (7), clip (6), seed (10): generate noise; scale to $\mathrm{SNR}_{\mathrm{in}}$; $x = s + n$
Causal filtering $y_c = H\{x\}$ (single pass)
Delay compensation: lag $\tau = \arg\max$ xcorr$(s, H\{s\})$
Zero-phase offline filtering y_z (forward–backward, needs future samples)
Compute $\mathrm{SNR}_{\mathrm{out}}$, ΔSNR, MSE (+ output-referred and STFT-magnitude ΔSNR)
All trials done?
no
yes
Aggregate mean ± std; frame-by-frame streaming test vs. batch; timing
Tables, figures, paper